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PPP application in infrastructure development in China: Institutional analysis and implications

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Abstract

Public–private partnership (PPP) has been applied more and more widely for the past two decades. Questions still remain as to how to cultivate a facilitating institutional environment for developing PPP projects success. This paper examines such a question, focusing on the institutional analysis of the Chinese context. A theoretical framework of PPP governance is first developed, identifying the logics and interactions of the differentiating levels in the PPP system. Then, the institutional framework of China is presented with regard to its cultural, legal, and administrative characteristics, as the embeddedness of PPP development. Thereafter, the evolution of Chinese institutions with particular regard to PPP is analyzed. Findings show that changes in the institutional arrangements have to go in tandem with introduction of PPP, and performance of PPP is closely related with its institutional environment. Suggestions are put forward to enhance China's institutional arrangements for healthy PPP promotion, and the implications for governments and investors in other countries are described.
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Keywords: Public–private partnership; Infrastructure; Institutional arrangements; China

1. Introduction

Infrastructure is considered to be "life supporting services" (Noel and Brzeski, 2005) and has long been under the monopoly of government. However, insufficient infrastructure has been a problem for almost all countries; this problem is exacerbated by the lack of funds available in the public sectors (Matos-Castaño et al., 2014; Qiu and Wang, 2011). Further, the monopoly of government over infrastructure has been seen as a cause of low efficiency in the development and operation of infrastructure facilities. Public–private partnership (PPP), as an innovation of project delivery alternatives, has been widely applied in many countries in the past two decades, attempting to resolve the problems (Chan et al., 2011); now it is, arguably, the most significant, worldwide trend in the public sector (Garvin and Bosso, 2008; Koch and Buser, 2006), and good governance is an important factor for the success of PPP projects in terms of developing sound economic policy and administrating projects (Li et al., 2005).

Over the past two decades, the Chinese government has been embarking on an ambitious programme of large investments on infrastructure development. To facilitate urbanization in China, the funds required for urban infrastructure development during the first 20 years of the twenty-first century are expected to be around 3500–5000 billion RMB (Wu, 2007). Funds from government alone are unlikely to be available to finance such large investments and so, reforms need to be undertaken by the Chinese government regarding the investment and financing of infrastructure projects. PPP was thus introduced in China to alleviate this problem (Chan et al., 2009; Feng and Luo, 1999; Ge and Zhang, 2009; He, 2001; Wang, 2006). In fact, PPP, in its modern form, was applied in China in the late 1980s, first in industrial development projects, and later other sectors, particularly infrastructure. According to

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the World Bank (2013),[1] the total number of infrastructure projects with private participation from 1990 to 2012 reached 1064 in China, ranking first worldwide, with a total investment commitment being 119,330 million US dollars, ranking fourth worldwide, after Brazil, India and Russian Federation.

China is characterized by unique political, economic and cultural features (Buderi and Huang, 2006; Mu et al., 2011; Tan and Bian, 2013). Before the 1980s, China adopted a pure socialist economy, highly centralized and planned. Infrastructure investment and development was the sole responsibility of the government. With the introduction of PPP in its infrastructure development, and considering the complexity of such economic transactional activities, the PPP practice in China has become a focus of study by scholars in recent years, particularly with regard to PPP risks (*e.g.* Cheung and Chan, 2011; Ke et al., 2010; Song et al., 2013; Wang et al., 1999),potential problems(*e.g.* Chan et al., 2010a), critical success factors (*e.g.* Chan et al., 2010b; Zhao et al., 2010) and case studies intending to draw lessons and experience from the implementation of real PPP projects (e.g., Chen, 2009; Chen and Hubbard, 2012). These studies offer us a better understanding of the current status, problems and constraints encountered in PPP projects in China. They also suggest or imply that sound institutions are very important for PPP success and that there is a need for China to create a PPP-enabling institutional environment (Wang et al., 2012). However, our literature review shows that there still exists a lack of systematic institutional analysis, especially from the dynamic evolution perspective, to address the following question: how have China's institutional arrangements evolved to accommodate the new project financing and delivery approach in infrastructure development?

This paper, with a view to contributing new knowledge on PPP administration, conducts an in-depth analysis from an institutional perspective. It is structured as follows: Section 1 is introduction. Section 2 presents a theoretical framework for guiding the analysis by taking the perspective of the new institutional economics. Section 3 provides an in-depth analysis of the evolution and changes in the institutional arrangements by investigating the development trajectory of PPP in infrastructure in China. Section 4 concludes with the findings identified from the analysis and presents topics for further studies with regard to governance issues of PPP.

2. Developing a theoretical framework

2.1. Conceptual issues of PPP

The literature offers many definitions of PPP and debates over how the term should be defined (Ball, 2011); it is even suggested that trying to define such a term is of little use (Hall et al., 2003). This phenomenon can be explained, at least partially, by the fact that PPP is a vague construct that is used by the international community. The term PPP is of a very general nature, with various origins from different countries, regions, and organizations, for example, private finance initiative (PFI) in UK, build–operate–transfer (BOT) in Asia and Pacific Regions, concession in France, private participation in infrastructure (PPI) of the World Bank, each with possible relevance to their political, economic, legal, cultural, and industrial backdrops. These terms have gradually converged to the term of PPP. As Grimsey and Lewis (2005: p. xiii) put it, "… the PPP is evolving in different ways in each country in which the arrangements are being implemented".

Definitions of PPP can be offered with different perspectives. For example, FHWA (2004: p. viii) describes it as "… a [long-term] contractual agreement formed between public and private sector partners, which allows more private sector participation than is traditional. The agreements usually involve a government agency contracting with a private company to renovate, construct, operate, maintain, and/or manage a facility or system...., ". Similar 'definitions' are given from the perspective of the contractual arrangement (*e.g.* Infrastructure Australia, 2008; Yong, 2010). After a critical review of current literature on defining PPP, Brinkerhoff and Brinkerhoff (2011) alleged that, function-specific definitions of PPP (*e.g.* Grimsey and Lewis, 2004, 2005; Koppenjan, 2005), which focus on provision of infrastructure financing, construction, operation and maintenance, are of little help in delineating the key features of PPP. Instead, they propose that mutual commitment between the public and private organizations is the key feature of PPP. They recognize two dimensions of PPP: "mutuality" and "organization identity" in the cross-sectoral collaboration. A delving analysis of PPP definitions, in addition to the above, suggests three general categories: (a) contractual-perspective (*e.g.* Bult-Spiering and Dewulf, 2006; Garvin, 2010), (b) partnership-focused (*e.g.* MOF, 2012; Nijkamp et al., 2002) and (c) function-specific (*e.g.* Grimsey and Lewis, 2004; IMF, 2006). The contractual perspective and partnership-focused perspective see PPP from the points of view of two different types of "relationship". The contractual perspective focuses mainly on the formal and legal dimensions of the relationship that binds the government and private partner together while the partnership-focused perspective mainly emphasizes the social dimension of the relationship, characterized with mutual commitment and trust. The function-specific perspective is task-oriented and takes PPP as an integration of all the fragmented work roles throughout the whole project cycle, including financing, design, construction and operation, all transferred from the government to the private partner, particularly the shift of the financing and operation roles from government to private partner. However, these three categories of definitions, though each focusing on a different perspective, can be seen as complementary rather than exclusive. Therefore, as an integrated perspective, PPP can be defined as a strategy that is utilized to provide quality infrastructure facilities and services with high efficiency (as purpose and function), based on a long-term contractual arrangement between public and private parties (as basis of formal governance of the relationship), through the synergetic cooperation between the public and private partners (as means and key feature).

Although the concept of PPP is not new in China, no official definition has been offered by the Chinese government. Practically, PPP in infrastructure in China often takes the

[1] see http://ppi.worldbank.org/.

specific forms of BOT, build–transfer (BT), transfer–operate–transfer (TOT), to name a few, with BOT being the dominant form (Adams et al., 2006). Very often, BOT is taken as the equivalent form of PPP, using the combined expression of "PPP/BOT" (Wang and Ke, 2008). There is one distinctive difference concerning the roles in the partnerships in China. "Public" in PPP in China refers to the government and "Private" refers to "enterprise", including both private enterprises and state-owned enterprises. This is because state-owned enterprises operate their business quite independently and across sectors. Therefore, the enterprise in a PPP project is deemed to be the "private party", even if it is state-owned. For example, in the Nanjing across the Yangtze River Tunnel Project, a typical PPP/BOT project in China, the public party is the Nanjing Municipal Government and the "private" party is China Railway Construction Corporation Ltd, which is a purely state-owned enterprise under the administration of the State-owned Assets Supervision and Administration Commission of the State Council of China. PPP in China, in a more strict sense, refers to Government–Enterprise Partnerships (GEP), and enterprises, more often than not, are state-owned. Thus, PPP, in the Chinese context, refers to the long-term cooperation between the government and enterprises that include both private and state-owned enterprises.

PPP, as a long-term contractual arrangement for the cooperation and coordination of government, investors, contractors, off-takers or end users, and all other stakeholders that are involved, has much relevance to both its institutional environment and organizational process throughout the whole development and operation cycle (Gopakumar, 2009; Urio, 2010; Winch et al., 2012; Zhang, 2005). Therefore, the governance issues of PPP are concerned with providing "rules for good order" at both institutional level (macro level) and organizational level (micro level). This paper focuses analysis of the institutional arrangements for PPP in China's context.

2.2. Theoretical framework

The whole PPP cycle generally runs through three major stages: development, realization, and operation. The development stage starts with the project initiation and ends at the financial closure or the signing of the concession contract. The realization stage comprises design development, construction, and commissioning. The operation stage is relatively simple as compared to the realization stage, but lengthy, thus exposed to external uncertainties along within the remaining concession period which is usually fifteen to thirty years. This PPP process assumes that nothing goes wrong in the interim and the project operates to the expectations of all participants.

A myriad of factors, including natural conditions, institutions, modes of organization, game rules, market conditions, technology, and history, influence the outcomes of PPP projects (Chen and Doloi, 2008; Urio, 2010; Zhang, 2005). A simple framework is useful as it isolates key variables and provides a clearer view of fundamental forces underlying specific empirical situations that might, otherwise, be overlooked. Therefore, a framework is developed to delineate the analytic boundary and guide the institutional analysis of PPP in China.

In trying to develop such a framework, a proper way is to base it on the very fundamental and time-honored theories. For the purpose of an institutional analysis of the transaction of PPP projects, the new institutional economics (NIE) perspective is adopted in this paper. According to Williamson (2000), four levels of factors matter for economic transactions. The first level, which refers to embeddedness, consisting of informal institutions, customs, traditions, norms and religion, acts as constraints for the transactions. Factors on this level are often taken as non-calculative and spontaneous, and indirectly influence the outcome of transactions through the second level: formal institutions, which embody the formal rules, such as the proper rights, polity, judiciary and bureaucracy which are reflected in the legal and administrative systems. This level is intended to get the institutional environment right for economic transactions. The third level is governance, i.e., the playing of the "game". The main purpose of this level is to align governance structures with transactions and, if the right structure is designed and utilized, transaction costs can be reduced or avoided. This level deals with the direct organization of transactions with contract design as the lens. The fourth level is resource allocation and employment that concerns prices, quantities and incentive alignment. This level aims to get the marginal conditions right. As between the four levels, each higher level imposes constraints on the lower level from Level 1 to Level 4; and each lower level gives feedback to each higher level from Level 4 to Level 1.

Based on the above concepts and taking the whole PPP transaction into context, a framework is developed to act as the lens for the institutional analysis of PPP project transactions, as shown in Fig. 1.

The PPP project process, from a micro perspective, mainly concerns resource allocation and employment (Level 4 in Fig. 1), which is the focus of project execution and management (intra-organizational). This execution process is formally governed by the contractual rules designed to regulate the participants (inter-organizational). In turn, these contractual arrangements are constrained by the institutional environment, both informal and formal, through which they operate. The informal institutions include the perceptions of PPP by the general public and a PPP-facilitating or impeding culture, while the formal institutions include constitution, laws, regulations, decrees, directives and administrative structures of the government. The institutional environment functions as constraints or facilitators that influence the PPP transaction process directly or indirectly. As can be seen from Fig. 1, the initial development stage of the whole PPP process is mainly governed by institutional rules. The path for developing a PPP project at this stage follows the legal framework and the administrative rules, with underlying latent practices dictated by culture (informal institutions), as, at this stage, the contracts that govern the specific relationships of the PPP participants are not yet in place. The realization and operation stages are formally governed by contractual relationships that regulate everyday operation of the project which are nested in a hierarchy of the

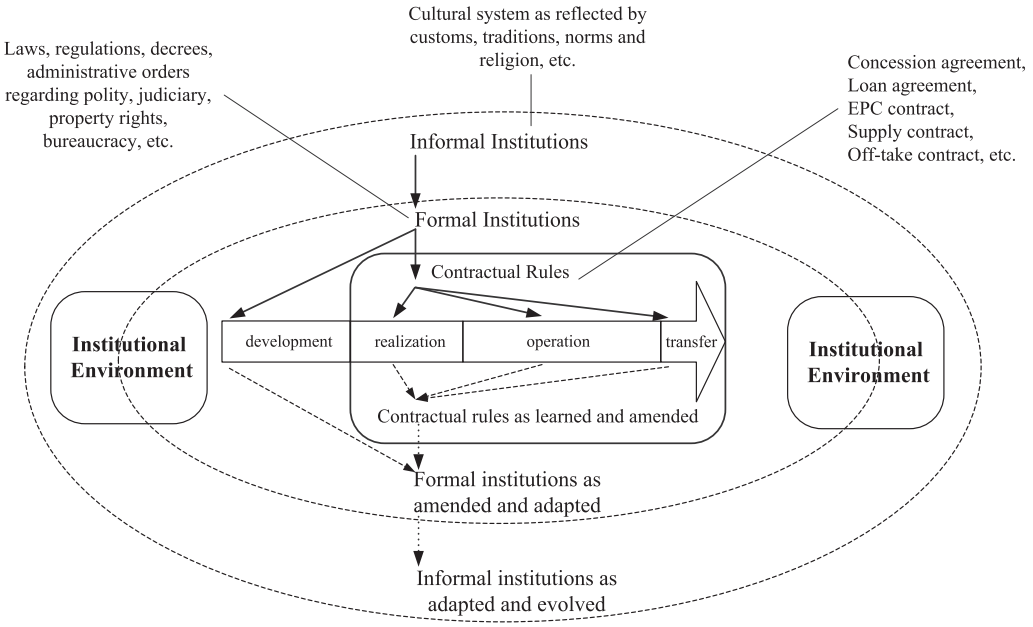


Fig. 1. Institutional framework for PPP development.

institutional environment. In addition, feedback emerges from Level 4 to Level 1, as a process of organizational learning, given the fullness of PPP project life cycles. The solid arrows represent the influence from a higher level to a lower level and the dashed arrows signal feedback from a lower level to a higher level. Such relationships are displayed in the organization networks constituted by concession contracts, shareholders' agreements, financing agreements, construction contracts, supply contracts, off-take agreements. The institutional and contractual rules, with their interactions with each other, constitute the governance system of the whole PPP project.

Guided by the above framework, the following sections analyze the development of PPP in China's infrastructure sector, focusing on its institutional context, i.e., Level 1-informal institutions and Level 2- formal institutions.

3. Development of PPP in China: institutional analysis

3.1. The general institutional framework in China

Institutions are the norms and rules of a society and affect the performance of economies by reducing uncertainty in guiding human interactions and structuring incentives in such exchange (North, 1990). These institutional rules are rooted in and reflected by the cultural system (informal institutions), and the legal and administrative systems (formal institutions).

3.1.1. Chinese cultural system

The Chinese culture is traditionally rooted and dominated by Confucianism, characterized with the "Doctrine of the Mean", which sets great store by equal or average distribution of wealth rather than wealth production itself (*bu huan pin er huan bu jun,* emphasis on *"end" rather on "means"*) (Peng, 1999; Zhang, 2004). This mindset has been aggravated since 1949 when the socialist new China was founded. The Chinese cognitive style in processing information and making judgment is, generally, in the order of "*heqing, heli, hefa*" (emotionally sensible, cognitively reasonable, and legal) in handling social affairs (Fellows et al., 2011; Peng, 1999). This mindset remains very strong among the general public, although it is alleviated to some extent, due to China's reform and opening-up policy, as an attempt to restructure its institutional arrangements, to call for individual initiatives and encourage economic development. This collective mentality not only affects the enacting of laws but also the implementation of the PPP projects, where private participation in the infrastructure and public utilities suggests "privatization" in the eyes of the public and arouses anti-PPP sentiments. For example, the Beijing International Airport Toll Express Highway, a PPP (concession) project, has been a prime topic of public debate in the past several years, when its excessive toll charges and exorbitant profit was disclosed by the press. There have been repeated calls from the public for cessation of its toll charging. Heavily pressured by the public, the government and the project investor had to renegotiate and finally reached a compromise that the toll rate be halved from July 1, 2011.[2]

Guanxi (personal ties/relationships) and authority are also distinctive features of Chinese culture. People tend to seek help from those with whom they have *guanxi* in social and business affairs. It is often believed that, to be successful in business, formal relationships should be blended with personal ones (Buderi and Huang, 2006). In order to obtain a favorable position in the competition, it is very important for PPP bidders to build *guanxi* with the key government officials who have authority and could offer support for winning the project.

[2] see http://www.chinanews.com/auto/2011/07-28/3217964.shtml.

However, unlike common business transactions, it should be kept in mind that, for such a long-term transaction as a PPP project, too much dependence on *guanxi* with government officials is not advisable. Some *guanxi* may not be effective (Su et al., 2007) and may even pose political risks. For example, in a solid waste power generation project in which one of the authors got involved, the project bidder won the concession contract with the support of the city mayor. However, at the early stage of the project implementation, the city mayor left for a new position and the project came to a halt due to the land acquisition problem because the new mayor was not in favor of this project and did not give full support.

3.1.2. General structure of the Chinese legal system

China has a unique and complex legal system (Zhu, 2013; Zhu and Han, 2013). Traditionally, it was based on the civil law system but recently it has been absorbing principles gradually from the common law system, and characterized by socialist ideology. It consists of a large number of laws which, in a broad sense, can be classified into a number of levels: cardinal law, fundamental and general laws, bye-laws and regulations, and provisions or rules (Huai, 2012; Tan and Bian, 2013). The structure of these levels of laws is shown in Fig. 2.

The cardinal law at the top level is the Constitution and is laid down and promulgated by the National People's Congress. The second-level laws cover fundamental and general laws. Fundamental laws include civil, contractual, property, administrative, criminal and procedural laws, also promulgated by the National People's Congress. The general laws, made by the Standing Committee of the National People's Congress, cover such laws as the Bidding Law, the Government Procurement Law, the Building Law, the Environment Protection Law. The third level is administrative regulations, made by the State Council. The fourth level covers the provincial bye-laws and administrative rules made by the ministries of the State Council. Rules of Provincial governments and bye-laws of major cities (as approved by the State Council) belong to the fifth level. The sixth level, the lowest level, involves rules of major city governments. Each higher level takes precedence over lower levels. The administrative rules made by ministries and the provincial bye-laws are of the equivalent level, but apply to their respective domains. Likewise, the bye-laws of major cities and the rules of provincial governments are equivalents. Although the higher-level laws take precedence over lower-level laws theoretically, the lower-level laws function and operate as guides in reality, as they tend to be more specific and concrete and are easy to follow.

3.1.3. General framework of the Chinese administrative system

The State Council, representing the People's Central Government, is the highest administrative authority. As the leader of the all administrative ministries and organizations countrywide, it is, among other things, in charge of planning national economic activities, leading and managing urban and rural construction, and allocating administrative functions to the provincial local governments. With regard to infrastructure, several ministries and administrations are specifically relevant, including the National Development and Reform Commission, the Ministry of Finance, the Ministry of Housing and Rural–Urban Development, the Ministry of Commerce, etc.

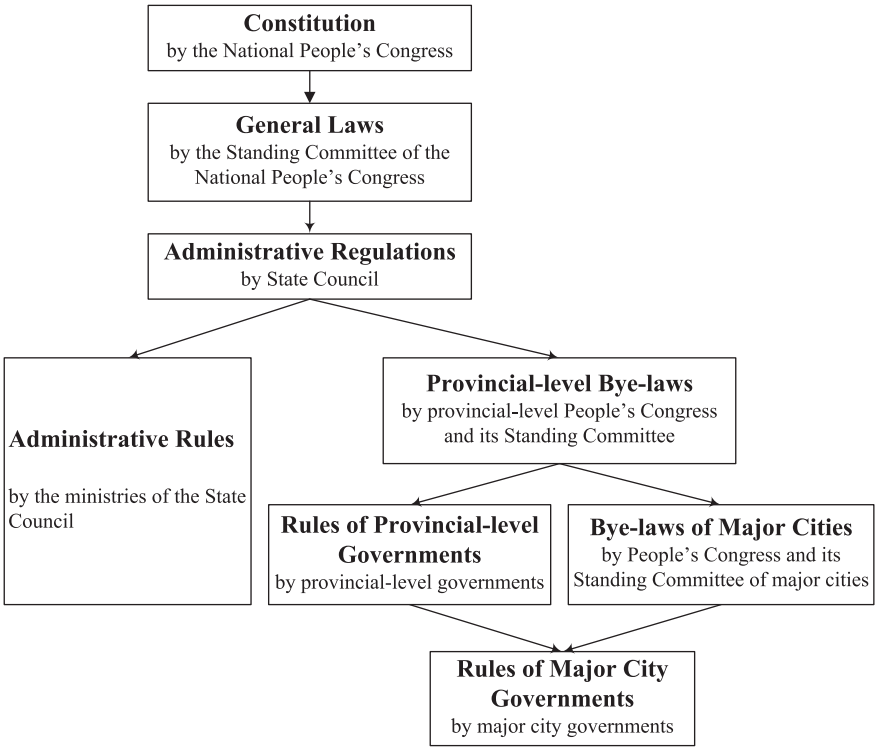


Fig. 2. Levels of laws in the Chinese legal framework.

Currently in China, all the various infrastructure sectors are governed dually by local governments in *block* (horizontally) and the ministries of the State Council in *line* (vertically) (Jumbo, 2007; Wang, 2006).

The framework of the administrative structure is shown in Fig. 3.

3.2. The evolution of institutions with specific regard to PPP

Against the backdrop of the introduction of China's Reform and open-up policy in the early 1980s, the development of PPP in China has gone through two major phases:

3.2.1. The pilot phase (1983–1999)

During this period, China was in great need of funds to develop its national economy; both public and private sectors were short of money. The open-up policy led to the establishment of many Sino–foreign joint ventures, mainly in industrial projects in the 1980s and gradually expanded into infrastructure development from early 1990s. To go with the application of PPP projects and attract foreign investors, policy documents were promulgated by governmental departments. "The Circular Concerning Relevant Issues in Attracting Foreign Investments by Means of BOT"(*Waijingmao Fa Han Document No.89*, 1994) was issued by the then responsible Ministry of Foreign Economic and Trade Cooperation of China (MFECT) in 1994, as a policy call with the main purpose of attracting foreign investment in infrastructure. A follow-up document, entitled "The Circular on Several Issues Concerning Review, Approval and Administration of Pilot Foreign-invested Concession Projects", was issued jointly in 1995, by the then State Planning Commission (SPC, predecessor of the National Development and Reform Commission), the then Ministry of Electric Power (later decomposed into several major state-owned electric power enterprises) and the Ministry of Transportation. This document, again, encouraged the adoption of BOT mode for infrastructure development and particularly for transportation projects. More benign conditions were offered to foreign investors in BOT projects.

A small number of PPP projects in the form of BOT were introduced and implemented, formally or informally, during this period. Shenzhen Shajiao B Power Plant in Guangdong Province was initiated in 1984, and was regarded as the first pioneering BOT project with the investor being a Hong Kong based company. Due to lack of experience on the part of the Chinese government, the government bore excessively lopsided risks for this project (Wang and Ke, 2008). Five hallmark infrastructure projects were formally selected by the SPC as pilot projects during this period. Of all the pilot projects, only two sustained the development and survived, i.e., Laibin Power B Plant in Guangxi Province and Chengdu Sixth Water Plant in Sichuan Province, despite the lengthy pre-contract negotiations and onerous coordination between various governmental departments, central and local, across a number of sectors. The others were implemented unsuccessfully, and Wuhan Junshan Yantze River Bridge Highway Project in Hubei Province, as one of the selected pilot BOT projects, was never formally initiated before it came to a full stop (Jin, 2005). The first wave of PPP projects in China gradually receded with the occurrence of the 1998 Asian financial crisis, coupled with the numerous coordination problems across governmental departments and sectors, encountered in the implementation of the BOT projects.

3.2.2. The development phase (2000–2012)

Drawing experiences and lessons from the previous pilot PPP projects, special regulations and policies (See Table 1 below) were made by the Central Government (the State Council) and

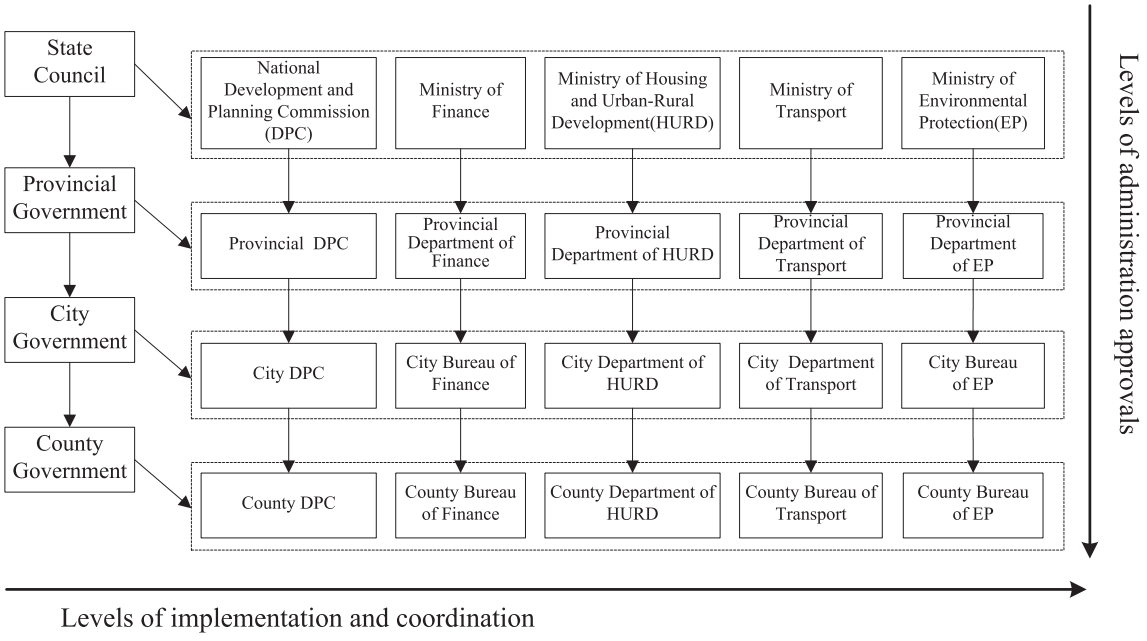


Fig. 3. The general administrative framework of China.

relevant ministries to lead, encourage and regulate the rapid development of PPP. During this phase, more and more public sector projects came to be open to private investors. Following the policies of the Central People's Government, local governments at various levels issued policies in their administrative regions. Thereafter, more and more PPP projects have emerged. One of the recent salient examples of PPP projects in China is the main stadium of Beijing 2008 Olympic Games, the Birds Nest, for which the National Stadium Co., Ltd (Project Company, established jointly by the Beijing State-owned Assets Management Co., Ltd. and CITIC Group) was granted the concession to finance, build, and operate the Bird's Nest and transfer it to Beijing City Government at the expiration of the concession period. Other PPP examples are Line 4 of Beijing Subway Project and Line 4 of Shenzhen Subway Project.

The two waves of PPP development in China are depicted in Fig. 4, with regard to numbers of PPP projects from 1990 to 2000, and from 2001 to 2012, respectively in four sectors.

A list of key legal and policy documents promulgated in China with specific regard to PPP up to date is shown in Table 1.

3.2.3. Institutional transition and PPP performance

It can be seen from the above that the introduction of the PPP projects at the initial phase was of a pilot nature, and the main purpose was to attract foreign investments to improve the then badly needed infrastructure facilities particularly in energy/power and transportation sectors, as the domestic private economy at that time was immature and very weak, and could not afford the project financing. Accordingly, efforts in trying to introduce new policies or to change the existing rules were made by the government (mainly at the ministry level) and focused on foreign investments, as institutional rules to govern this new mode of project development. As a result, the number of PPP projects generally increased for the first few years. However, the performance of the first-stage projects was, generally, far from satisfactory with a large proportion of project failures (Wang, 2002). The policies did attract PPP investments quickly; however, they did not guarantee the sustainability of such projects, as these policies were of low legal-order and quite general. Besides, the administrative structure of the government, central and local, did not adapt to the new situation by restructuring its administrative departments, but remained unchanged, and exercised tight control, leading to conflicts across the relevant sectoral government departments. Taking advantage of the general policies of the Central Government, local governments, with a simple view to attracting more PPP projects in a shorter term and due to lack of experience, offered excessively favorable terms to the project investors. They signed unbalanced concession agreements by bearing too many risks and, accordingly, could not keep their promises during the execution of the projects, resulting in defaults by local governments and disputes between them and the investors, with the PPP project Case of Shenyang No.9 Water Supply Plant being a good illustration (Qi et al., 2009). The chaotic conditions created by the pilot PPP projects led the Central Government to issue notices to "clear up" such PPP projects, leading to the decrease in numbers of PPP projects at the end of the 1990s for several years. The policies at the pilot stage were just a reflection of the political will, characterized with project development expediency as a showoff of "outstanding performance" of the

Table 1
Key policies and regulations with specific regard to PPP promulgated.

Year	Promulgator	Name of the Law or Regulation
1994	MFECT[1]	The Circular Concerning Relevant Issues in Attracting Foreign Investments by Means of BOT
1995	SPC[2], MOEP[3], and MOT[4]	The Circular on Several Issues Concerning Review, Approval and Administration of Pilot Foreign-invested Concession Projects
1997	SPC and FEA[5]	The Provisional Administrative Method for Project Financing Obtained from Overseas
2002	MOC[6]	The Circular on Accelerating the Liberalization of the Urban Utilities Market
2002	SPC and MOC	The Circular on Promoting the Industrialization of Urban Wastewater and Solid Waste
2004	MOC	The Administrative Method of Urban Utilities Concessions
2004	State Council	Decisions on the Reform of Investment Mechanisms
2005	State Council	Opinions of Encouraging, Supporting and Introducing Private Economy Development
2007	NDRC[7]	The Amended Foreign Investment Sectors Catalogue
2010	State Council	Directives of Promoting and Guiding Healthy Development of Private Investment
2011	NDRC	The Circular on Promoting Development of Strategic Rising industries for Private Enterprises
2012	MHRUD	The Circular on Further Promoting Private Investment in Urban Public Utilities
2012	NDRC	The Circular on Price Leverage to Promote Private Investment
2012	NDRC *et al.*	Advice and Guidelines for Private Oversees Investment
2013	State Council	Advice on Renovating Investment and Finance Mechanisms in Railroad Projects
2013	State Council	Advice on Enhancing Urban Infrastructure
2014	NDRC	Guidelines for Natural Gas Infrastructure: Construction, Operation, and Management

[8]MHRUD: the Ministry of Housing and Rural-Urban Development.
[1] MFECT: Ministry of Foreign Economic and Trade Cooperation of China.
[2] SPC: the State Planning Commission.
[3] MOEP: the Ministry of Electric Power.
[4] MOT: the Ministry of Transport.
[5] FEA: State Administration of Foreign Exchange.
[6] MOC: Ministry of Construction.
[7] NDRC: National Development and Reform Commission.

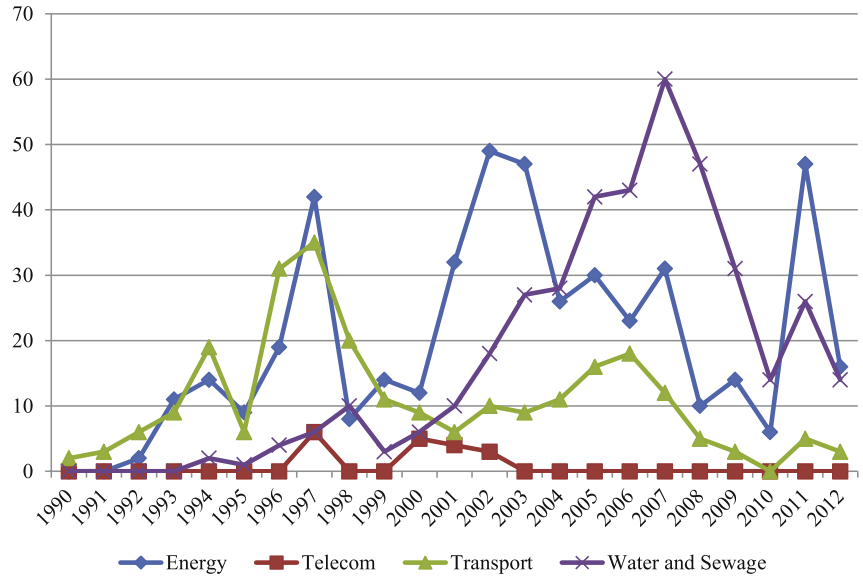


Fig. 4. Nos of PPP projects in China from 1990 to 2012.

governmental officials. However, this project development expediency was severely constrained by the poor institutional environment, notably improper then existing regulations and even conflicts between the policies of the Central Government and local governments, often leading to PPP implementation organizations not knowing how to proceed and, eventually, to unsatisfactory outcomes. A survey of this phase showed one of the major risks for PPP investors was political risk due to lack of a sound PPP governance system (Wang et al., 2000).

The development stage since 2000 has witnessed continued efforts by government to promote PPP projects in China. The Amendment of the Constitution which was passed in 2004 by the National People's Congress, makes a further step in protecting private property rights, by clearly specifying that "the State protects the legal rights and benefits of non-stated-owned businesses, including individual business and private business… the State encourages, supports and directs the development of the non-state-owned economies…(Article 21)" and "…private properties shall not be infringed (Article 22)", thus, endeavoring to create a benign formal environment for private investment. The PPP mode is even directly specified in the Highways Law, passed in 2004 by the NPC Standing Committee, in that "the State encourages domestic and foreign economic entities to invest in the construction of highways (Article 21)" and that "the toll

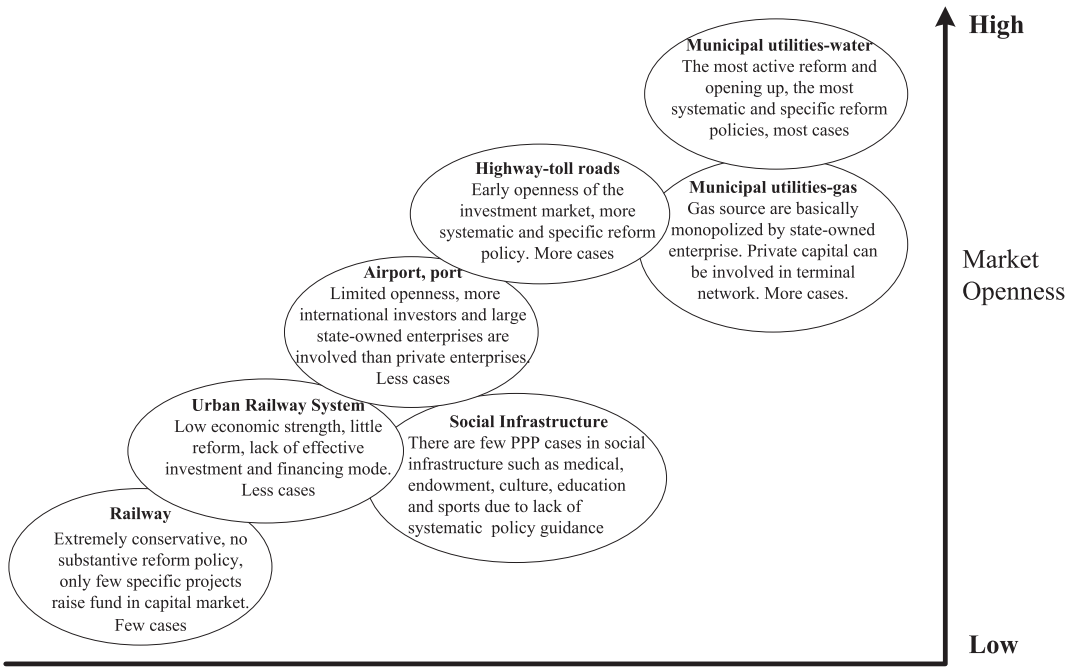


Fig. 5. Current PPP sector market openness in China.

highways invested by domestic or foreign economic entities, shall, upon expiration of the concession period, be transferred to the State without being charged fees, which process shall be administered by the competent departments of transportation (Article 65).” Three administrative regulations were issued consecutively by the State Council during this stage and are seen as hallmark policies for further reform of the China infrastructure investment system; the main purpose is to enhance the efficiency in providing infrastructure. The first one is entitled “Decisions on the Reform of Investment Mechanisms [*Guofa* 2004, no 20]” that makes clear calls for diversifying investment channels and attracting private funds through the market system to invest in infrastructure projects and public utilities by means of PPP. In doing so, government administrative functions shall be restructured to serve such a reform of investment system better. The second one, entitled “Opinions of Encouraging, Supporting and Introducing Private Economy Development [*Guofa* 2005, no 3]”, aims to relax the market entry conditions for the private investors, offers support to them in terms of financing and taxes, nurturing and developing professional service providers to PPP projects, etc. A more recent document entitled “Directives of Promoting and Guiding Healthy Development of Private Investment [*Guofa* 2010, no 13]” is the most updated policy in further promoting the implementation of PPP. This document

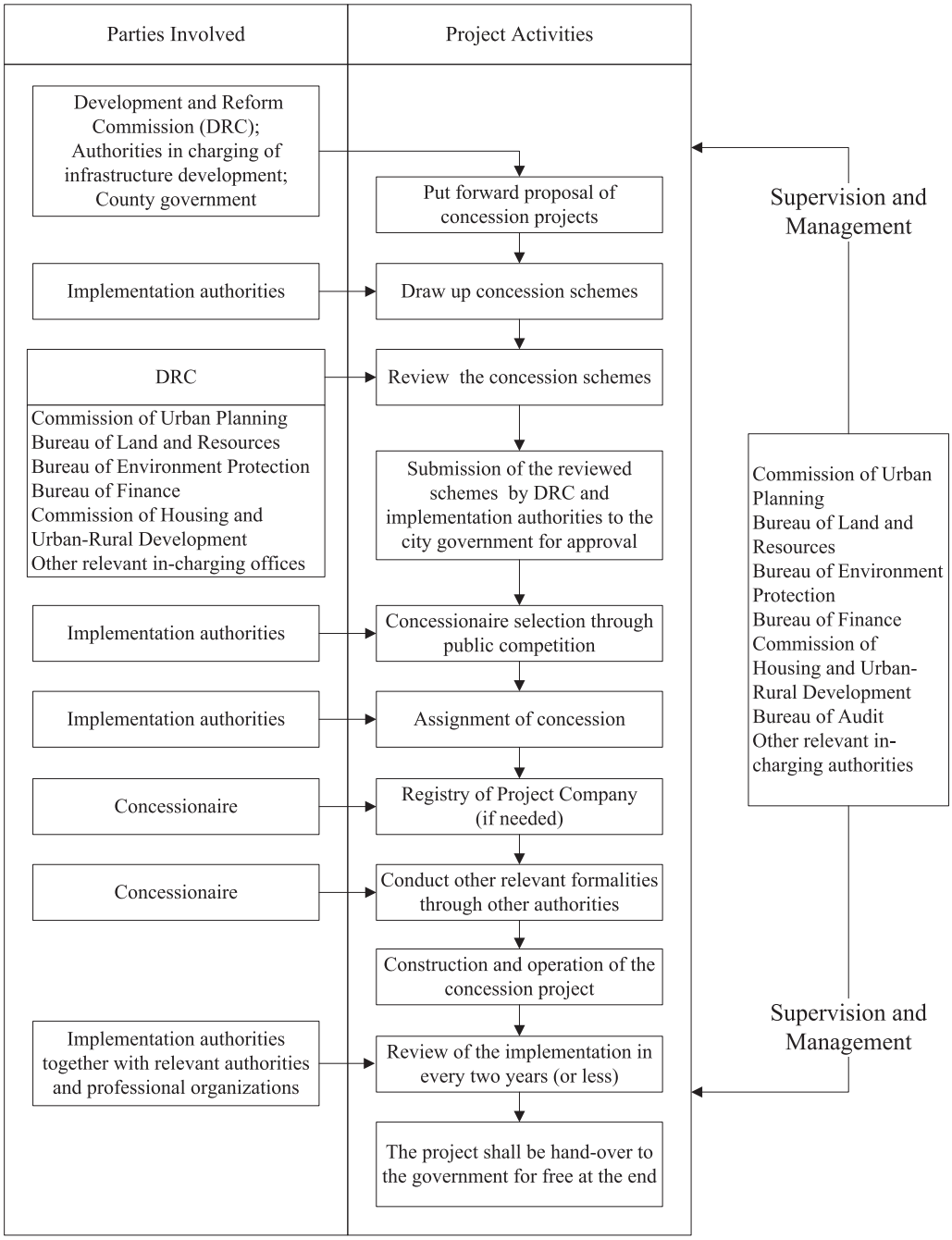


Fig. 6. Administrative structure for PPP project implementation in China.

requests all levels of government to provide an equal investment opportunity for private investors. It clearly opens markets in almost all infrastructure sectors, such as transport, energy (electrical power, oil and natural gas), land reclamation and mining, telecommunications, and even railway (a sector under the monopoly of government up to date), encouraging private investors to participate in various PPP forms. To go with the policies of the State Council, various ministries also released specific rules to govern PPP transactions. For example, the Ministry of Construction (predecessor of the current Ministry of Housing and Urban-Rural Development) published PPP documents to act as specific guides for PPP implementation in its administration domain. The Ministry of Construction, in addition to releasing its document "The Administrative Method of the Urban Utilities Concessions (2004)", prepared and released five model PPP/concession agreements respectively for water sector, pipelined gas supply, urban refuse treatment, urban heating supply. The current market openness is depicted in Fig. 5.

Under the framework set by the State Council and ministries, more and more bye-laws and rules at the provincial or ministerial level have been laid down. According to incomplete statistics by the authors, 46 cities, including Shanghai, Tianjin and Chongqing, have released similar legal documents. Based on these legal and policy documents, the administrative structure specifically for PPP project implementation has evolved into tentative shape. Depending on the scope and location of the project, a PPP project can be proposed or initiated either by the Development and Reform Commission (DRC), or authorities of the infrastructure development for a specific sector (such as department of transport) at each level of the administrative structure. Thereafter, an implementation authority or execution organization will be set up to act as the granter (project owner) on behalf of the government, and is put in charge of all pre-contract activities, including preparing the detailed PPP scheme, conducting feasibility study, and reporting to DRC and relevant departments for review and approval. Once approved, the project will be implemented in a step-wise manner. The implementation authority, with the coordination and interaction with other governmental departments, will be in charge of selecting the project sponsor (concessionaire), managing the design, construction, operation and transfer stages, to ensure the intended performance outcome. The resulting PPP administrative framework is shown as in Fig. 6.

3.2.4. Towards a maturity PPP phase as an outcome of institutional change

Although efforts have been made by the Chinese Government for the past 20 years to try introducing PPP in infrastructure, what's behind its efforts has been a test-see-learn-and-do philosophy (incremental change), intending to avoid unaffordable costs for potential failures.

Before the introduction of the PPP approach, the Chinese institutional environment for infrastructure development was generally stable. Such stability of the institutional framework had been accomplished by a complex and highly-coupled set of constraints that included both formal and informal rules in a hierarchy which had evolved to meet the traditional infrastructure project development approach, typically the design-bid-build, where the government provided all the project funds and operation service. The resulting contradiction between an urgent need for much more infrastructure and severe shortage of funds lured it to change to the PPP approach. Such a change of preference by the government acts as an exogenous change agent, making it necessary to adjust the then stable, complex and highly coupled institutional system. The process of institutional change has followed an incremental pattern and is still evolving to a new institutional equilibrium, i.e., a maturity phase. As shown in the above analysis of the two PPP development phases in China, the change was first initiated by introduction of lower-level policies (quick-responsive, short-time oriented and flexible), and then new formal rules by higher authorities, followed by change of administrative structures which are based on the new formal institutional rules (relatively stable). As indicated in Fig. 1, the change of the informal institutions (Level 1 in Fig. 1) is much slower and harder than that of the formal institutions (Level 2 in Fig. 1). The institutional change pattern in China echoes with the proposition that the persistence of cultural traits in the face of change of formal rules makes informal constraints change at a different rate from formal rules (North, 1990). The institution-evolving route for PPP in China is depicted in Fig. 7.

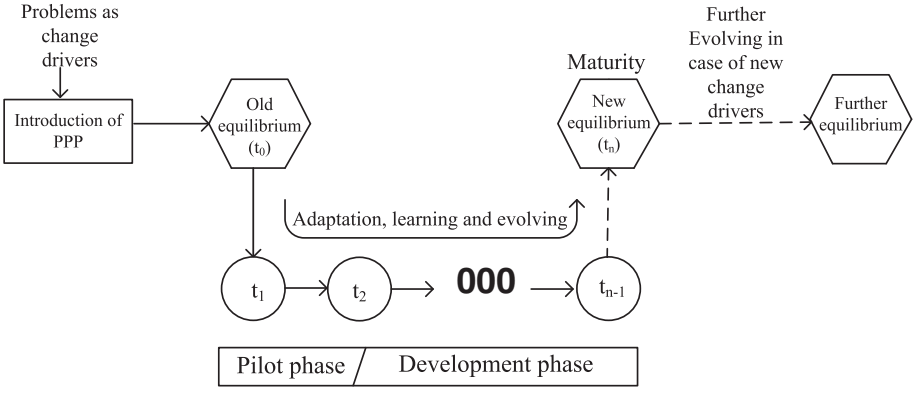


Fig. 7. Institution-evolving route for PPP in China.

4. Conclusions and suggestions

This paper develops a theoretical framework for PPP development, followed by an intuitional analysis of PPP governance within the Chinese context. Findings from the analysis indicate that the introduction of PPP, as a new transaction approach in infrastructure, proceeded in tandem with changes of the institutional arrangements, and performance of PPP models is closely related to its institutional environment, formal and informal. Piece-meal policies and documents issued by the Chinese government did attract a number of PPP investors. Such policies and documents, however, could not guarantee the sustainability of PPP projects. At the present stage, the institutional environment for PPP in China is still far from mature, particularly the administrative structure. The administrative structure for PPP development, though evolving, still acts as a negative constraint. Government departments, at levels of line ministry and local governments across all the infrastructure sectors, have discrete functions and none of them want to lose their administrative power or lessen control. Problems thus occur when PPP projects go through the administrative procedures. For example, in a recent case of a light-rail PPP project that had been approved by the City Development and Reform Commission(Grantor on behalf of the City Government), the project sponsor was allowed to develop commercial properties affiliated to this light-rail project, as a financial subsidy to the Sponsor. However, when the Sponsor designed the project with office buildings on top of the station terminal for sale as means for financing the project, the Urban Planning Bureau called for a stop of this activity in that the land approved for the project could be used for the rail project only. The discrete functions for the project under the current administration structure constrained p'roject innovation. This is consistent with Sha's (2004) findings that control and intervention from government departments, especially by local government officials, is the major source of problems within China's construction industry which increases transaction costs in China's construction market.

Very recently the government has been continuing to create legislation and policies to enhance the institutional environment and to regulate the behavior of governmental departments. From 2012 to early 2014, new implementation rules and guides were released by relevant ministries and city governments to materialize the general policy document "*Directives of Promoting and Guiding Healthy Development of Private Investment*" issued by the State Council in 2010, as further promotions of private/enterprise participation in infrastructure development (For details, see Table 1 in Section 3). A key in promoting PPP in China is to iron out the kinks in the institutional environment by deregulating vertical approval, enhancing horizontal coordination across governmental departments, and cultivating a benign culture. As a matter of fact, scholars and practitioners have been aware of the urgency for the Chinese government to establish a sound legal and institutional system for successful PPP application in China (Peng, 2011; Wang and Zhang, 2013). To fill the gaps of the current institutional arrangements for PPP, further efforts should be made by the Chinese government with regard to the following:

- Amend the current laws and regulations which are in contradiction or conflict with the PPP model, and prepare and put into effect a unified and higher-order PPP/BOT Law, instead of continuing to issue piecemeal measures.
- Under the framework of PPP/BOT law, prepare and release a unified PPP development procedure guide by the State Council; based on which, detailed guides will be developed by local governments, who are the actual PPP execution organizations, for each infrastructure sector.
- Establish independent regulators as a third party to supervise the whole project cycle and gather feedback information as learning and fertilization across various PPP sectors.
- Set up PPP units within the Government organizations as the special PPP execution vehicle to facilitate the development process and PPP knowledge accumulation such that institutional capacity for PPP execution can be enhanced.
- Establish a public participation mechanism in selecting PPP projects, by means of which the public, as a stakeholder, is entitled to participate in the hearings, thus creating a democratic and transparent PPP project selection process, to facilitate and nurture a positive perception and recognition of PPP.

These suggested reforms of the current institutions cannot be achieved unless the key stakeholders, with the government being the dominating actor, form a concordant mental map and translate such a uniform mentality into collective actions as a learning outcome of the PPP development and implementation. This is by no means an easy task, as interest-vested actors (again notably some governmental departments who will not be willing to lose their power in the administrative approvals) in the current institutional system will likely exert resistance to such change.

The findings from this paper also have implications for governments of other countries. The impact of institutional environment on PPP performance cannot be overemphasized by governments, and therefore they should cultivate benign institutional environments to match the PPP approach if good performance is to be achieved. The test-see-learn-and-do philosophy adopted for PPP application in China has been shown to be an effective method that is worth learning by other countries, as institutional change is a slow and sensitive process, and often subject to resistance. It would incur heavy and sunk costs, particularly for infrastructure development, if done inappropriately. However, other countries should also learn from the lessons China suffered in introducing PPP. For example, as discussed in Section 3, the Chinese government, out of its political will, issued quite a number of piecemeal policies for promoting PPP at the early stage, but did very little to restructure its administrative departments to go with the policies, and this resulted in a number of practical problems. Also, the lack of high-order PPP laws failed to guarantee the sustainability of PPP projects, as indicated by the high project failure rate.

For investors from other countries intending to bid for PPP projects in China, three aspects should merit their attention. First, the lack of high-order and unified PPP law requires that private investors should, as a complement, try their best to sign a more detailed and complete concession contract, clarifying the roles and responsibilities as between the investor and the government; however, it is not advisable for investors to attempt to sign a risk-lopsided contract in their favor by taking advantage of the government officials' lack of experience, as this might be perceived later to be an unfair deal which is most likely to lead to renegotiation or even contract abandonment. Secondly, investors should be aware of the complex but sometimes ambiguous relationships between the relevant administrative departments, which may require a large input of effort in coordination with relevant departments during PPP project implementation because China is a country where governments are still very strong and play dominating roles in economic activities. A well-designed concession contract is necessary, but coordination with governments is more important. Finally, investors should not rely too heavily on their personal ties to win the contract, although China is believed to be a *guanxi*-oriented country. Government officials might help investors to win contracts. However, since PPP projects usually last as long as fifteen to thirty years and officials' positions are short term, a change of official during the project would probably put the investors in an unfavorable position and continual support from new officials might be in question. It is suggested that they, instead, should be more concerned about the project viability, with regards to its economic, social and environmental aspects.

This paper focuses its analysis on the institutional level (particularly the formal level); however, good governance of the PPP concerns, equally if not more, the organizational level, which exerts more direct influence on the outcome. As shown in Fig. 1, the "playing of the PPP game" involves a large number of "actors" across the whole project cycle that may last more than 30 years. Designing the "rules of the game" (contract) at the organizational level and exploring the interactions between institutional level and the organizational level may offer more in-depth insights. Therefore, further studies of PPP governance are suggested to focus on this integrative across-levels analysis, preferably by systematic, longitudinal multi-case studies. These studies particularly should concern such topics as (a) the impact of institutional environment on the concession contractual arrangements; (b) the learning and response effects from each lower level to each higher level (i.e., from Level 4 to Level 1 as in Fig. 1); (c) developing PPP institutional maturity assessment models; (d) the effects of perceived unfairness of concession contract on PPP performance; and (e) the coupling effects between contracts in the contractual network of PPP project, considering the interweaving roles each contract plays along the complex PPP project "supply chains". Once such dynamics between the institutional layers are identified and such maturity models are built, better strategies could be developed for the sustainability of PPP projects and, thus, a sound PPP governance system will be created.

Conflict of interest statement

The authors declare that there are no conflict of interest.

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